Naïve Bayes? Not so Naïve for Predicting Term Deposit Subscriptions (June 2024)

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ABSTRACT Predicting customer behavior in finance, particularly their propensity to subscribe to term deposits, is crucial for informed marketing strategies. This work explores Naïve Bayes classifiers, a probabilistic approach based on Bayes' theorem, for this task. We delve into the core principles of Naïve Bayes, including the concept of conditional independence and its impact on classification. This report utilizes a standard banking dataset to train and evaluate Naïve Bayes models for predicting term deposit subscriptions. We explore the performance of different Naïve Bayes variants offered by scikit-learn to identify the most effective model for this specific dataset. Furthermore, to potentially enhance predictive power, the investigation goes beyond a single classifier by exploring the implementation of a hybrid approach that combines Naïve Bayes to handle both categorical and numerical data.

INDEX TERMS Bayes' Theorem, Classification, Conditional Independence, Hybrid Approach, Probabilistic Approach

I. INTRODUCTION

Naïve Bayes classifiers are a family of probabilistic algorithms widely used for various classification tasks, including customer behavior prediction in the financial sector. Their popularity stems from their efficiency, interpretability, and strong performance across diverse datasets.
Probabilistic Foundation:
Naïve Bayes relies on Bayes' Theorem, a fundamental concept in probability theory. This theorem allows us to calculate the "posterior probability", which is the probability of a specific class (e.g., subscribing to a term deposit) occurring given certain features (e.g., income, age, investment history). By calculating this posterior probability for each class, Naïve Bayes assigns the data point to the class with the highest probability.
Conditional Independence Assumption:
The "naïve" aspect of Naïve Bayes comes from its key assumption: features are "conditionally independent" of each other given the target class. In simpler terms, the presence or absence of one feature doesn't influence the presence or absence of another feature, considering the class label. While this assumption might not always hold true in reality, it simplifies the model and allows for efficient calculations. For example, imagine predicting whether a customer will subscribe to a term deposit. Features might include income, age, and existing investments. Naïve Bayes assumes that knowing a customer's income doesn't directly influence their age or investment habits, considering their subscription decision.
Advantages of Naïve Bayes are as follows:
Efficiency: Training Naïve Bayes models is computationally efficient compared to more complex algorithms like Support Vector Machines or Neural Networks. This makes them suitable for real-time applications and processing large datasets.

Interpretability: Unlike some "black box" models, Naïve Bayes allows an understanding of how features contribute to the final prediction. By analyzing the posterior probabilities for each feature value, we can gain insights into which features are most influential for a specific class. This interpretability is crucial in financial settings, where understanding the rationale behind customer behavior predictions is important.
Scalability: Naïve Bayes performs well with large datasets due to its efficient nature and ability to handle high-dimensional data (many features).
Handling Missing Values: Naïve Bayes can handle missing data points relatively well using techniques like imputation (estimating missing values based on other features). This is advantageous in real-world scenarios where data might not be complete.
Limitations of Naïve Bayes are as follows:
Naïve Assumption: The assumption of feature independence can lead to inaccurate predictions if features are highly correlated. In real-world datasets, features often exhibit some degree of correlation. This limitation can be mitigated through techniques like feature selection (choosing only the most relevant features).
Sensitivity to Noise: Naïve Bayes can be sensitive to noisy data, potentially impacting accuracy. Outliers or irrelevant features can skew the calculation of posterior probabilities. Techniques like data cleaning and outlier detection can help address this limitation.
Class Imbalance: Naïve Bayes might struggle with imbalanced datasets where one class significantly outnumbers others (e.g., a small percentage of customers subscribing to term deposits).

II. RELATED WORK

Predicting customer behavior in the financial sector, particularly their propensity to subscribe to specific products, has been a growing area of research. Machine learning algorithms have emerged as powerful tools for this purpose, with several studies exploring their effectiveness in predicting term deposit subscriptions.
One common approach involves using classification algorithms. For example, Mendes-Filho et al. [1] compared the performance of various algorithms, including Naïve Bayes, Decision Tree, Random Forest, Support Vector Machine, and Neural Networks. Their findings suggest that while Naïve Bayes achieved good accuracy, algorithms like Decision Trees offer slightly better performance.
Another study by Freitas et al. [2] specifically focused on Naïve Bayes classifiers for predicting term deposit subscriptions. They investigated the impact of the "naïve" feature independence assumption and employed Laplace smoothing to address potential issues with sparse data. Their results indicated that Naïve Bayes could achieve good accuracy.
Furthermore, Ferreira et al. [3] explored a wider range of algorithms on the same Portuguese Banking Dataset used in this study. Their analysis confirmed that while Naïve Bayes performs well, Decision Trees with pruning techniques might offer superior results in terms of accuracy.
These studies highlight the potential of machine learning algorithms, particularly Naïve Bayes, for predicting term deposit subscriptions. They also emphasize the importance of comparing different algorithms and tailoring the choice based on the specific dataset and desired performance metrics.

III. METHODOLOGY

A. DATASET DESCRIPTION

Naïve Bayes classification is a probabilistic machine learning algorithm based on Bayes' theorem, assuming independence between features. It is commonly used for binary and multi-class classification tasks due to its simplicity and efficiency, providing robust performance even with small datasets.
The Bank Marketing dataset used for the experiment consists of data about a Portuguese banking institution's direct marketing campaigns performed through phone calls with no missing values. The objective is to predict whether or not the client will subscribe to a term deposit which is represented by target variable 'y' containing binary values 'yes' and 'no'.
The dataset consists of 16 attributes and 45211 instances with each instance representing client information including demographic details, financial status, contact history, and campaign interactions. Since multiple contacts can be made to the same client, multiple instances can represent the same client at different times.
The dataset is stored in a CSV file and contains the following attributes:

1. age: This attribute is an integer data type that represents the age of the client
2. job: The job attribute is a categorical data type with 12 categories: admin, blue-collar, entrepreneur, housemaid, management, retired, self-employed, services, student, technician, unemployed, and unknown representing the profession of the client.
3. marital: The marital attribute represents the marital status of the client represented in categorical form with 4 categories: 'divorced', 'married', 'single', and 'unknown' where the 'divorced' category includes divorced and widowed individuals
4. education: The education attribute represents the highest level of education acquired by the client. It is a categorical data type with 8 categories: basic.4y, basic.6y, basic.9y, high.school, illiterate, professional.course, university.degree, unknown
5. default: This attribute indicates whether or not the client has credit in default using binary data type.
6. balance: The balance attribute, represented using integer data type, contains the average yearly balance of the client.
7. housing: This attribute indicates whether or not the client has a house loan using binary data type.
8. loan: This attribute uses binary data type to indicate whether or not the client has a personal loan
9. contact: The contact attribute represents the communication type, i.e., cellular or telephone using the categorical data type
10. day_of_week: The day_of_week attribute represented using the date data type indicates the last contact day of the week.
11. month: This attribute represents the last contact month of year
12. duration: This attribute contains the last contact duration in seconds in integer form. The duration attribute highly affects the target and should be included for benchmark purposes and should be discarded for a realistic predictive model
13. campaign: This attribute represents the number of contacts performed to that particular client during this campaign in integer form
14. pdays: This attribute represents the number of days that passed after the client was last contacted from a previous campaign where -1 indicates that the client was not previously contacted.
15. previous: This attribute represents the amount of time this particular client was contacted before this campaign using an integer value.
16. poutcome: The poutcome attribute is represented using 3 categories: failure, nonexistent, and success to represent the outcome of the precious marketing campaign.

B. PROPOSED METHODOLOGY

The 'Bank Marketing' dataset used in this report contains categorical and integer attributes and discrete outcomes making it ideal for classification tasks. The Naïve Bayes Classification can be applied to this dataset using two methods: converting all the integer attributes to categorical and converting all the categorical attributes to integer form.

The computation of Naïve Bayes Classification involves several steps.
Before commencing the classification task, first of all, data preprocessing has to be done. It involves server key steps:
First, the quality of the dataset was assessed. For this, missing values, data redundancy and outliers were checked. Since the Bank Marketing dataset does not contain missing values, data preprocessing techniques like imputation or dropping features or rows were not needed to be performed. There was no case of da-ta redundancy either. So, there was no need for detecting and removing the redundant data.
However, feature importance calculation should be performed using correlation analysis to identify irrelevant features and those attributes are dropped in order to improve the model’s performance.
For fitting Gaussian Naïve Bayes classifier, all the features should be numerical. So, all the integer attributes are converted to categorial form and the categorical data is then converted into numerical labels using one-hot encoding. Furthermore, the numerical features have to be normalized. For this, standard scaler is used, which scales

normalizes the data within the values ranging from 0 to 1. Also, the binary type data is also encoded using 0 and 1, where 0 represents negative class and 1 represents positive class. Before splitting the dataset into train set and test set, the dataset is checked for imbalanced dataset and the target class is stratified during the train-test split in case of im-balanced dataset. For balanced datasets, the dataset is then split into train and test data without stratifying the target class. After that, the training set is used for training the GaussianNB classifier.

Similarly, the process is repeated for Categorial Naïve Bayes Classifier, where all the numerical data is converted to categorial form. For this, we discretize the continuous features data. This can be done using two techniques, namely Equal width discretization and Equal frequency discretization. Using equal width discretization, the continuous data is distributed in bins having equal intervals. For example, one bin contains data from 1 to 10, then second bin contains data from 11 to 20, then third bin contains data from 21 to 30 and so on. In case of equal frequency distribution, the continuous data is distributed into bins such that each bin contains about same amount of data. For example, one bin contains 20 instances, then all other bins should also contain about 20 instances. After discretization of the continuous data, all the categorical data is encoded using label encoder. Finally, the dataset is split into train set and test set, checked for imbalanced dataset, and stratified accordingly. After that, the training set is used for training the CategoricalNB classifier.

GaussianNB is suitable for numerical data and CategoricalNB is suitable for categorical data. However, our dataset contains both numerical and categorical data. So, we need a hybrid Naïve Bayes Classifier for our dataset. Since there is no in-built package for this, we prepared the hybrid Naïve Bayes by ourselves. This hybrid model leverages the functionalities of GaussianNB for dealing with continuous numerical data and that of CategoricalNB for dealing with categorical data. The final result is obtained by summing the log probabilities of the predictions from both the model.

Once trained, the model is evaluated using accuracy, precision, recall, and F1-score. Confusion matrices and histograms are used to visualize the model's performance.

Algorithm: Naïve Bayes Classification

1. Import necessary libraries
2. Load data from CSV file
3. Perform data preprocessing
4. For GaussianNB, transform categorical features to numerical features and for CategoricalNB, discretize continuous features
6. Perform feature importance analysis
7. Ensure conditional independence
8. Handle class imbalance in dataset
9. Split the data into training and test sets
11. Train decision tree classifier using the training set using GaussianNB classifier for numerical data and CategoricalNB classifier for categorical data
12. Predict target variables for the test set
13. Evaluate classifier performance

C. MATHEMATICAL FORMULAE

The Naïve Bayes classifier trained used the following formulas:

Naïve Bayes Formula: The Naïve Bayes classifier calculates the probability of a class label C whose input features x are given. The Naïve Bayes classifier assumes conditional independence between features of a given class label.

$$P(C|x) = \frac{P(x|C) \times P(C)}{P(x)} \tag{1}$$

where, $P(C|x)$ is the probability of class C given the features x, $P(x|C)$ is the probability of features x given class C, $P(C)$ is the prior probability of class C and $P(x)$ is the probability of features x.

The formulae used for performance evaluation are:

Accuracy:

$$Accuracy = \frac{TP + TN}{\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN}} \tag{2}$$

where, TP (True Positives) represents correctly predicted positive instances, TN (True Negatives) represents correctly predicted negative instances, FP (False Positives) represents incorrectly predicted positive instances and FN (False Negatives) represent incorrectly predicted negative instances.

Precision:

$$Precision = \frac{TP}{\mathrm{TP} + \mathrm{FP}} \quad (3)$$

Recall:

$$Recall = \frac{TP}{\mathrm{TP} + \mathrm{FN}} \quad (4)$$

F1-score:

$$F1\ Score = 2\mathrm{x}\frac{Precision * \ Recall}{\mathrm{Precision} + \mathrm{Recall}} \quad (5)$$

where, Precision is the ratio of correctly predicted positive instances to the total predicted positive instances and Recall ratio represents the correctly predicted positive instances to the total actual positive instances.

D. INSTRUMENTATION DETAILS

The analysis and visualization in this study were conducted using Jupyter Notebook, a versatile and interactive web-based coding environment that allows for the integration of code execution, rich text, mathematics, plots, and rich media in a single document. This platform was chosen for its ease of use and its extensive support for data science and machine learning workflows.

The core libraries used in this study were NumPy, Pandas, Scikit-learn, Matplotlib, and Seaborn. NumPy is a fundamental library for scientific computing in Python, providing support for large multi-dimensional arrays and matrices, along with a collection of mathematical functions to operate on these arrays. In this study, NumPy was used for various numeric operations, such as array manipulations and mathematical computations that are integral to the preprocessing and analysis of data.

Pandas is essential for data manipulation and analysis, offering data structures and functions needed to manipulate structured data seamlessly. Pandas was utilized for importing the dataset, handling missing values, and performing Exploratory Data Analysis (EDA). The library's DataFrame structure was particularly useful for organizing the data and preparing it for analysis.

Scikit-learn, a robust machine learning library in Python, provides simple and efficient tools for data mining and data analysis. It was used extensively in this study. The train_test_split function was employed to split the dataset into training and testing sets, essential for evaluating the model's performance. Preprocessing tools such as StandardScaler, OneHotEncoder, and LabelEncoder were used to normalize and encode the data. The GaussianNB and CategoricalNB classifiers from sklearn.naive_bayes were used to train the Naïve Bayes classifier models. Evaluation metrics including accuracy_score, precision_score, recall_score, f1_score, confusion_matrix, roc_curve, and auc were used to calculate and assess the performance of the models.

For data visualization, Matplotlib and Seaborn were employed. Matplotlib provides a comprehensive plotting framework, while Seaborn builds on Matplotlib to offer a high-level interface for drawing attractive and informative statistical graphics. These libraries were used to create visual representations of the confusion matrix, aiding in the interpretation of the classifier's performance, and to generate histogram plots to visualize the distribution of data and results.

IV. EXPERIMENTAL RESULTS

Feature Importance Analysis: Before performing the Naïve Bayes classification, a feature importance calculation was performed. The correlation of the features with the target variable was calculated and plotted as shown in Figure 1.

The features with the highest correlation with the target feature were duration, poutcome_success, poutcome_unknown, and contact_unknown. These features had correlation values higher than 0.15. Some other features, such as housing, month_mar, month_oct, month_sep, pdays, and month_may also had a somewhat high correlation (greater than 0.10).

Likewise, the features with the lowest correlation with the target features were job_unknown, job_selfemployed, month_aug, month_jan, and job_technician. These features had correlation values lower than 0.01. Most of the features had correlation of less than 0.10 with the target feature.

Check if the data is unbalanced: The dataset was found to be unbalanced, as illustrated in Figure 2. The number of instances of the negative class was approximately seven times that of the negative class. This indicates a very huge class imbalance, which can affect the performance of the Naïve Bayes Classifier. So, to mitigate this class imbalance, stratification was performed while splitting the data into training and testing sets. This helps to ensure that the class proportions are maintained in both the training and testing sets and reduces the risk of the model becoming biased towards the majority class.

Gaussian Naïve Bayes Classifier: A Gaussian Naïve Bayes Classifier was instantiated for numerical data of the dataset. For this, categorical variables were encoded with 1 for 'yes' and 0 for 'no'. One hot encoding was performed to convert categorical columns to numerical columns. A standard scalar was used for normalizing the numerical features. The Gaussian Naïve Bayes Classifier was trained on the training set and predictions were subsequently made on the testing set. The performance of the classifier was evaluated using a confusion matrix (Figure 3), a classification report (Table 1), a precision-recall Curve (Figure 4), and an ROC curve (Figure 5).

The TP (True Positive), TN (True Negative), FP (False Positive) and FN (False Negative) values for the Gaussian Naïve Bayes Classifier obtained from the confusion matrix were 766, 10853, 821 and 1124 respectively while the values for evaluation metrics accuracy, recall, precision and F1-score obtained from the classification report were observed to be 0.86, 0.91, 0.93 and 0.92 respectively.

The precision-recall curve for the Gaussian Naïve Bayes indicates that as recall increases precision gradually decreases suggesting that the true positive often comes at the cost of increasing false positives. The initial stagnant value of precision represents that the precision remains relatively constant with an increase in recall for a range of threshold values. This indicates the ability of the model to correctly identify additional positive instances without increasing the number of false positives. After a certain threshold, a slight increase in precision indicates a higher proportion of positive predictions being correct. However, there is a gradual decrease in the precision value towards the end indicating identification of more false positives thus reducing the precision.

Similarly, the ROC curve represents the ROC curve for class 1 represented by the orange line and ROC curve for class 0 represented by the green line. The area under the ROC curve for class 1 is 0.81 which indicates an 81% chance for the classifier to rank a randomly chosen positive instance higher than a randomly chosen negative instance indicating good performance. Since this curve lies towards the upper left corner, it indicates high TPR (True Positive Rate), proportion of positive instances correctly classified, and low FPR (False Positive Rate), proportion of negative instances incorrectly classified, which indicates that the model makes fewer errors in predicting the positive class. In the same way, the area under the ROC curve for class 0 is 0.19. Since the curve represents the ROC curve for binary classification, the sum of area under the curve adds up to 1.0 due to its symmetric nature. The blue dashed line in the ROC curve represents the performance of a random classifier having area under the curve value 0.5. For any classifier whose AUC is greater than that of the random classifier, will be represented in the upper left half, similar to the ROC curve of class 1, while that of the classifier having AUC less than 0.5 will be represented in the bottom right half, similar to the ROC curve of class 2. In summary, the curve indicates the ability of the model to distinguish between positive (class 1) and negative class (class 0).

Categorical Naïve Bayes Classifier: A Categorical Naïve Bayes Classifier was instantiated for categorical data of the dataset. For this, the numerical variables were discretized into 4 bins using Equal width and equal frequency discretization. Then, the categorical features were encoded using the Label encoder. The discretized and encoded datasets were split into training and testing sets, and the Categorical Naïve Bayes Classifiers were trained and evaluated.

Performance with Equal Width Discretization: The evaluation on equal width discretization was performed using a confusion matrix (Figure 6), a classification report (Table 2), a precision-recall curve(Figure 7), and an ROC curve (Figure 8).

The True Positive, True Negative, False Positive and False Negative values given by the confusion matrix were 533, 11474, 503, and 1054 respectively. Similarly, from the classification report, we observed that the model has an accuracy of 0.89 and precision, recall, and f1-score of 0.92, 0.96, and 0.94 respectively.

The precision-recall curve of the equal width discretized model indicates high initial precision values even at low recall values suggesting accurate positive predictions at predicting few positives. However, the curve shows a gradual decrease in precision with the increase in recall due to the presence of false positives.

Similar to the Gaussian Naïve Bayes Classifier, the orange line represents ROC curve of class 1, the green line represents ROC curve of class 0 and the blue dotted line represents the performance of a random classifier having area under the curve value 0.5. The AUC of class 1 is 0.78, therefore it is represented in the top left half of the blue dotted line representing high TPR and low FPR. The AUC represents how well the classifier ranks chosen positive instances over randomly chosen negative instances. Similarly, since the AUC of class 0 is 0.22, it is represented in the bottom right half of the blue dotted line representing low TPR and high FPR. Since this is a binary classification problem, the AUC remains symmetric and the sum adds up to 1.0.

Performance with Equal Frequency Discretization: Similar to previous models, the evaluation of the equal frequency discretization model was carried out using a confusion matrix (Figure 9), classification report (Table 3), precision-recall curve (Figure 10), and ROC curve (Figure 11).

The confusion matrix provided the following values: True Positives (TP) = 626, True Negatives (TN) = 11,485, False Positives (FP) = 492, and False Negatives (FN) = 961. According to the classification report, the model achieved an accuracy of 0.89, with precision, recall, and F1-score values of 0.92, 0.96, and 0.94, respectively.

The precision-recall curve for the equal frequency discretized model resembles that of the equal width discretized model with initial increase in precision and gradual decrease with an increase in recall.

Similarly, the ROC curve also resembles that of the equal width discretized model with AUC for class 1 being 0.87 and that of class 0 being 0.13. The curve for class 1 lies towards the top left half of the random classifier while the curve for class 0 lies towards the bottom right half of the random classifier. Since it is a single model binary classifier, the sum of AUC curve adds up to 1.0.

Hybrid Naïve Bayes Classifier: There is no built-in Hybrid Naïve Bayes Classifier. So, we created our own by combining Gaussian and Categorical Naïve Bayes Classifiers. The numerical features were trained on Gaussian Naïve Bayes and categorical features were trained on Categorical Naïve Bayes classifiers. For the aggregate result, the log probabilities obtained from both were summed after which the model was assessed using confusion matrix (Figure 12), classification report (Table 4), precision-recall curve (Figure 13) and ROC curve (Figure 14).

From the confusion matrix, the model showed 485 True Positives, 11,678 True Negatives, 299 False Positives, and 1,102 False Negatives. The classification report indicated an accuracy of 0.90, with precision at 0.91, recall at 0.98, and an F1-score of 0.94.

The precision-recall curve reveals that the model starts with high precision even at low recall levels, suggesting effective positive predictions when few positives are identified. However, as recall increases, precision steadily decreases due to the increase in false positives.

In the ROC curve, the orange line represents the ROC curve for class 1, the green line represents the ROC curve for class 0, and the blue dotted line denotes the performance of a random classifier with an AUC of 0.5. The AUC for class 1 is 0.85, indicating better performance than that of the random classifier placing it in the upper left half. This AUC reflects the classifier's effectiveness in ranking positive instances over negative ones. For class 0, the AUC is 0.13, indicating a low TPR and high FPR, placing it in the lower right half.

Condition of Conditional Independence: Naïve Bayes assumes the condition of conditional independence. To satisfy this, those features that have a low correlation with the target feature are dropped. Likewise, those features that have a high correlation with another feature are also dropped.

Columns dropped for having a low correlation with the target feature (threshold: abs(correlation) < 0.05) are:

['month_feb', 'education_secondary', 'month_jul', 'job_management', 'poutcome_other', 'day', 'job_services', 'age', 'default', 'job_unemployed', 'job_entrepreneur', 'month_jun', 'job_housemaid', 'month_nov', 'contact_telephone', 'education_unknown', 'month_jan' 'month_aug', 'job_self-employed', 'job_unknown', and 'job_technician',]

Then, columns dropped for having a high correlation among one another (threshold: abs(correlation)>0.5) are:

['pdays', 'previous']
Then, after satisfying the condition for conditional independence, the Naïve Bayes Classifier was instantiated and trained on the training set and predictions were subsequently made on the testing set. The performance of the decision tree was evaluated using a confusion matrix (Figure 15, Figure 18, Figure 21), a classification report (Table 5, Table 6, Table 7), a precision-recall Curve (Figure 16, Figure 19, Figure 22), and an ROC curve (Figure 17, Figure 20, Figure 23).
Initially, Gaussian Naïve Bayes Classifier was used by converting categorical attributes to integers while Categorical Naïve Bayes Classifier was used by converting integer attributes to categorical. Finally, a hybrid classifier was built by training numerical features using Gaussian Naïve Bayes Classifier and categorical features using Categorical Naïve Bayes classifier. The results of the two models were then aggregated by summing the log probabilities of both the models. All three models produced the same results.
The confusion matrix represented True Positive 1587, True Negative 11977, while both the False Positive and False Negative values were 0. Similarly, the classification report indicated the values of all four evaluation metrics, accuracy, f1-score, precision and recall, to be 1.0.
The precision-recall graph for the conditional model was that of a perfect model where the plot formed a square with a precision value of 1.0 throughout the entire range of recall indicating that every positive prediction made by the model was correct and the model produced no false positives, as shown by the classification report. This shows that the precision constantly remains at 1.0 as the recall increases from 0 to 1.
Similarly, the ROC curve of the model represented an ideal ROC curve for a binary classification model with two classes, class 1 and class 0. Class 1 has TPR of 1.0 at FPR of 0.0 while class 0 has TPR of 0.0 at an FPR of 1.0 forming a square which represents that the model correctly identifies all positive instances without false positives for class 1 and all negative instances without any false negatives for class 0.

V. DISCUSSION AND ANALYSIS

From feature importance analysis, we can gain key insights about the nature of the dataset. The feature with the highest correlation with the target variable is 'duration'. Duration shows a strong positive correlation with the target, suggesting that longer call durations are associated with a higher likelihood of the customer subscribing to term deposits. Additionally, 'poutcome_success' exhibits a notable positive correlation, indicating that a successful outcome from the previous marketing campaign significantly influences term deposit subscriptions. Conversely, 'poutcome_unknown' shows a strong negative correlation with the target, suggesting uncertainty in past campaign outcomes negatively impacts subscription rates.
Other features such as 'contact_unknown' and 'housing' (indicating whether the customer has a housing loan) also display significant negative correlations with the target variable. Furthermore, the analysis reveals that term deposit subscriptions tend to peak during the months of March, October, and September.
Among the features with a low correlation to the target, 'job' shows the least significant relationship, implying that occupation has minimal impact on term deposit subscription likelihood.
Let’s compare and analyze the performance of different Naïve Bayes classifier models used in this lab, namely Gaussian Naïve Bayes, Categorical Naïve Bayes, and Hybrid Naïve Bayes classifiers.
The Gaussian Naïve Bayes classifier achieved an accuracy of 86%, demonstrating solid overall performance. However, it is noteworthy that the precision, recall, and F1-score for the positive class were lower than those for the negative class. This imbalance can be attributed to the dataset containing fewer instances of the positive class, leading to a bias towards higher performance metrics for the negative class.
Moving to the Categorical Naïve Bayes classifier, we observed a slight improvement in accuracy to 89%. This model showed an increase in precision but a decrease in recall compared to the Gaussian Naïve Bayes classifier, resulting in a similar F1 score. Notably, the decision to employ equal frequency discretization contributed marginally to these improved results.
The Hybrid Naïve Bayes classifier, incorporating features from both Gaussian and Categorical models, achieved the highest accuracy among the three models at 90%. Despite this slight increase in accuracy, it exhibited a notable trend: the precision

remained relatively low while the recall was higher. This phenomenon suggests that the hybrid model may have improved its ability to correctly identify instances of the positive class (high recall) but at the expense of misclassifying some negative instances as positive (low precision).
The marginal improvement in performance despite using a hybrid model can be attributed to several factors. Both the Gaussian and Categorical Naïve Bayes classifiers already demonstrated strong performance, achieving accuracy rates exceeding 85% and F1-scores above 0.90 for the majority class. Consequently, the hybrid model faced limitations in further enhancing performance due to the dataset's inherent characteristics, including class imbalance and correlated features.
Let's delve into the significance of the Naïve Bayes assumption, i.e. the conditional independence assumption. Upon considering the conditional independence assumption for all Naïve Bayes classifier models, a profound impact on performance was observed. The models demonstrated flawless accuracy, precision, recall, and F1 scores, all reaching 100%. It posits that each feature in the dataset is conditionally independent of every other feature given the class variable. When this assumption holds true, Naïve Bayes classifiers can achieve optimal performance by effectively leveraging simple and efficient probabilistic computations. By assuming conditional independence, Naïve Bayes models become highly efficient in handling complex datasets with large feature spaces. This efficiency is evident in scenarios where the model correctly identifies and classifies instances with utmost accuracy, precision, and recall, leading to an overall F1 score of 1.00. However, it's essential to note that real-world datasets often deviate from perfect conditional independence.

VI. CONCLUSION

In conclusion, we explored Naïve Bayes classifiers as a probabilistic approach for predicting customer behavior in finance, specifically their propensity to subscribe to term deposits. We investigated the core principles of Naïve Bayes, particularly focusing on the concept of conditional independence and its impact on classification accuracy. Utilizing a standard banking dataset, we trained and evaluated various Naïve Bayes models to predict term deposit subscriptions. Our analysis revealed key insights through feature importance analysis, highlighting 'duration' as the most influential predictor positively correlated with subscription likelihood. Additionally, variables such as 'poutcome_success' and 'poutcome_unknown' provided critical insights into the impact of past campaign outcomes and uncertainties on subscription rates. Notably, 'contact_unknown' and 'housing' showed significant negative correlations with subscriptions, indicating their role in influencing customer decisions.
Comparing different Naïve Bayes classifiers, the Gaussian Naïve Bayes model achieved solid accuracy at 86%, albeit with imbalanced precision and recall metrics favoring the negative class. The Categorical Naïve Bayes model improved slightly to 89% accuracy, leveraging equal frequency discretization for enhanced performance. The Hybrid Naïve Bayes classifier outperformed both with 90% accuracy, demonstrating improved recall at the expense of precision in identifying positive class instances. Despite integrating features from both Gaussian and Categorical models, the marginal performance gain of the Hybrid Naïve Bayes model suggests limitations influenced by inherent dataset characteristics such as class imbalance and feature correlations. Moreover, our investigation affirmed the theoretical underpinning of the Conditional Independence Assumption, positing perfect accuracy, precision, recall, and F1-score metrics for Naïve Bayes classifiers under ideal conditions. This assumption simplifies complex data relationships, enabling efficient probabilistic computations for optimal model performance.
However, real-world datasets often deviate from perfect conditional independence, necessitating cautious consideration of model assumptions and dataset intricacies in practical applications. Future research could explore advanced techniques to mitigate these challenges and further enhance classification accuracy across diverse datasets, thereby advancing predictive capabilities in customer behavior analysis and informed marketing strategies.

APPENDIX

A. Feature Analysis

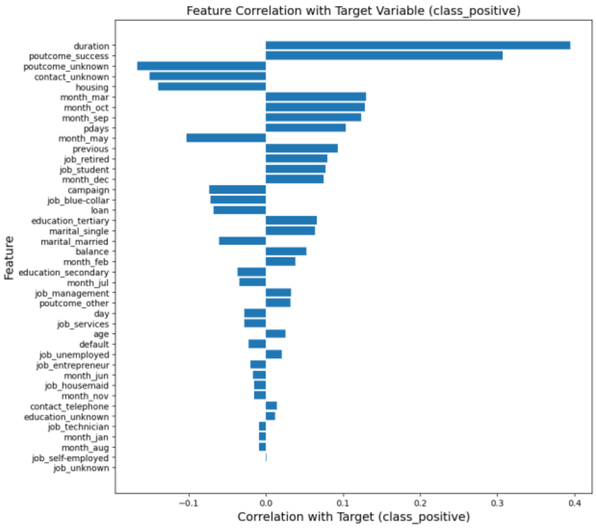

***Figure 1** Feature Correlation with Target Variable*

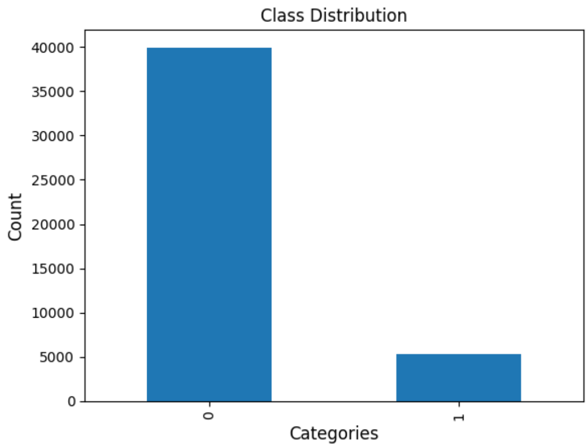

***Figure 2** Class Distribution*

B. Naïve Bayes Classification

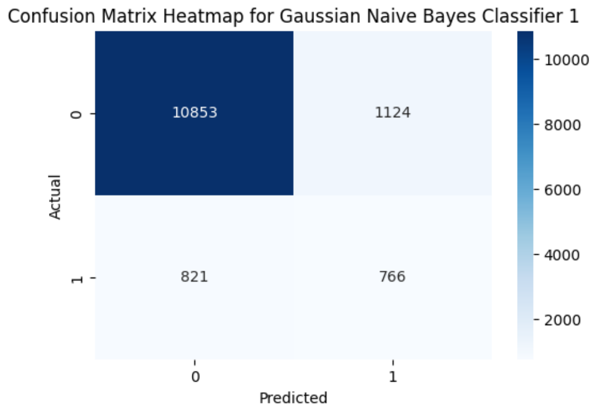

***Figure 3** Confusion Matrix for Gaussian Naïve Bayes Classifier*

***Table 1** Classification Report for Gaussian Naïve Bayes Classifier*

	Precision	Recall	F1-score	Support
0	0.93	0.91	0.92	11977
1	0.41	0.48	0.44	1587
Accuracy			0.86	13564
Macro avg	0.67	0.69	0.68	13564
Weighted avg	0.87	0.86	0.86	13564

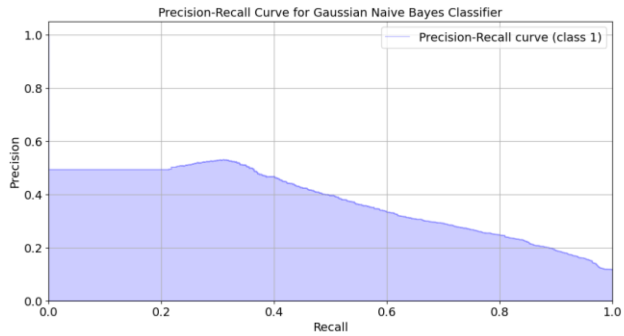

***Figure 4** Precision-Recall Curve for Gaussian Naïve Bayes Classifier*

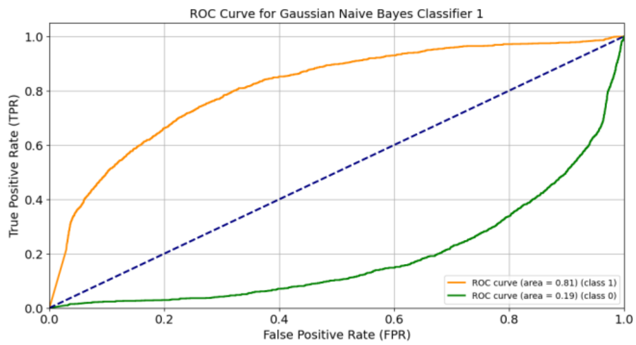

***Figure 5** ROC Curve for Gaussian Naïve Bayes Classifier*

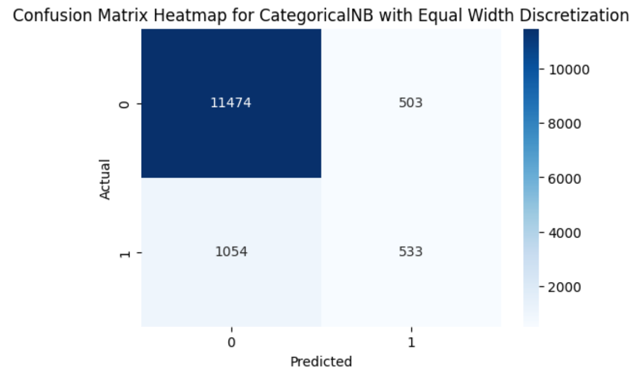

***Figure 6** Confusion Matrix for Categorical Naïve Bayes Classifier with Equal Width Discretization*

***Table 2** Classification Report for Categorical Naïve Bayes Classifier with Equal Width Discretization*

	Precision	Recall	F1-score	Support
0	0.92	0.96	0.94	11977
1	0.51	0.34	0.41	1587
Accuracy			0.89	13564
Macro avg	0.72	0.65	0.67	13564
Weighted avg	0.87	0.89	0.87	13564

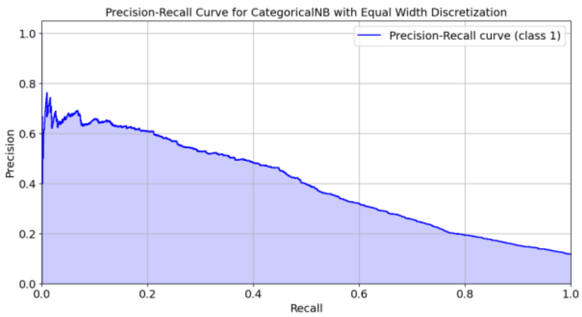

***Figure 7** Precision-Recall Curve for Categorical Naïve Bayes Classifier with Equal Width Discretization*

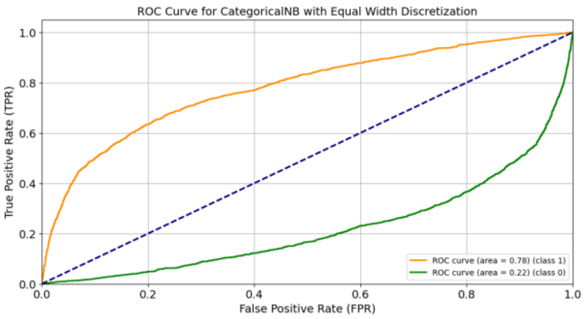

***Figure 8** ROC Curve for Categorical Naïve Bayes Classifier with Equal Width Discretization*

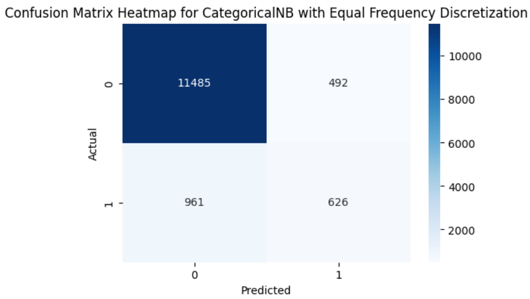

***Figure 9** Confusion Matrix for Categorical Naïve Bayes Classifier with Equal Frequency Discretization*

***Table 3** Classification Report for Categorical Naïve Bayes Classifier with Equal Frequency Discretization*

	Precision	Recall	F1-score	Support
0	0.92	0.96	0.94	11977
1	0.56	0.39	0.46	1587
Accuracy			0.89	13564
Macro avg	0.74	0.68	0.70	13564
Weighted avg	0.88	0.89	0.88	13564

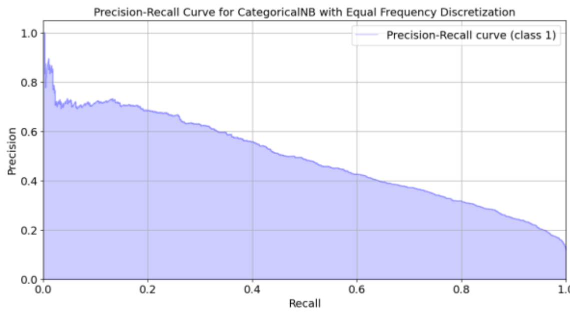

***Figure 10** Precision-Recall Curve for Categorical Naïve Bayes Classifier with Equal Frequency Discretization*

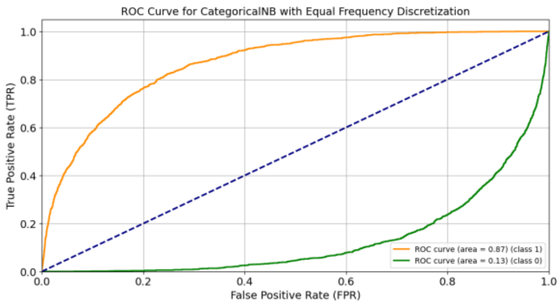

***Figure 11** ROC Curve for Categorical Naïve Bayes Classifier with Equal Frequency Discretization*

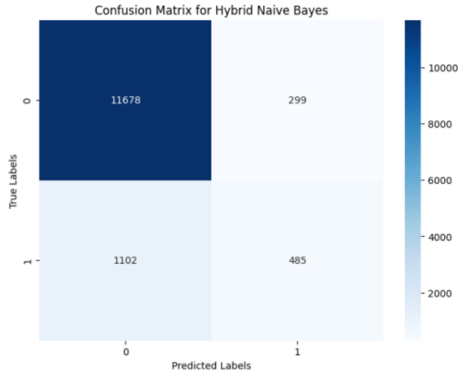

***Figure 12** Confusion Matrix for Hybrid Naïve Bayes Classifier*

***Table 4** Classification Report for Hybrid Naïve Bayes Classifier*

	Precision	Recall	F1-score	Support
0	0.91	0.98	0.94	11977
1	0.62	0.31	0.41	1587
Accuracy			0.90	13564
Macro avg	0.77	0.64	0.68	13564
Weighted avg	0.88	0.90	0.88	13564

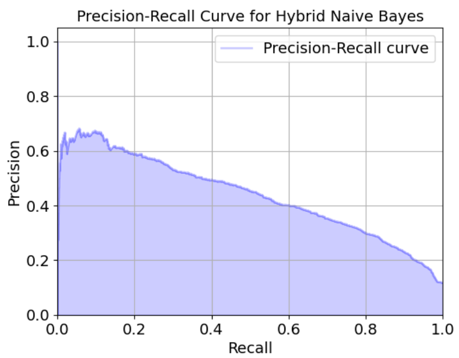

***Figure 13** Precision-Recall Curve for Hybrid Naïve Bayes Classifier*

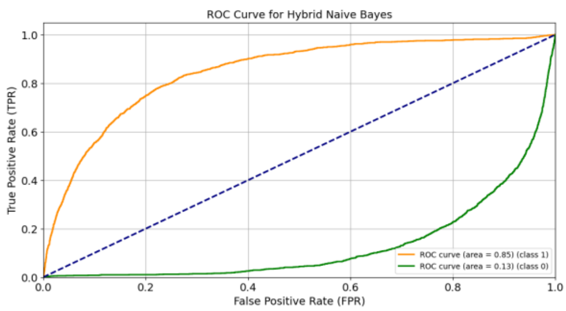

***Figure 14** ROC Curve for Hybrid Naïve Bayes Classifier*

C. After satisfying Conditional Independence Assumption

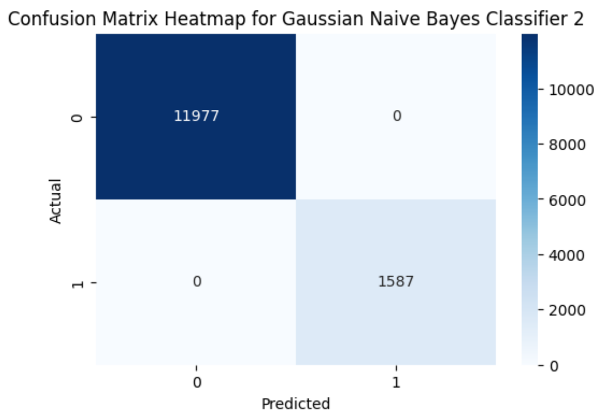

***Figure 15** Confusion Matrix for Gaussian Naïve Bayes Classifier*

***Table 5** Classification Report for Gaussian Naïve Bayes Classifier*

	Precision	Recall	F1-score	Support
0	1.00	1.00	1.00	11977
1	1.00	1.00	1.00	1587
Accuracy			1.00	13564
Macro avg	1.00	1.00	1.00	13564
Weighted avg	1.00	1.00	1.00	13564

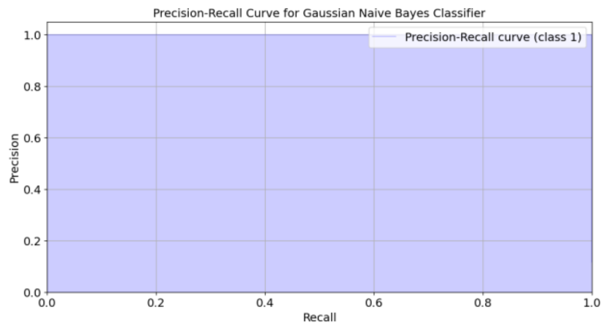

***Figure 16** Precision-Recall Curve for Gaussian Naïve Classifier*

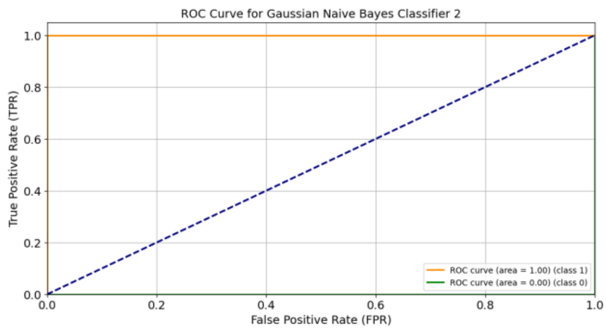

***Figure 17** ROC Curve for Gaussian Naïve Bayes Classifier*

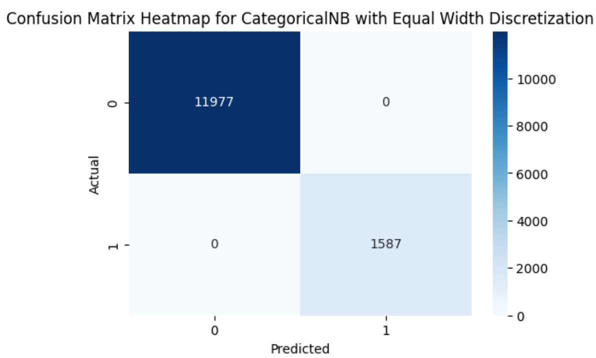

***Figure 18** Confusion Matrix for Categorical Naïve Bayes Classifier with Equal Width Discretization*

***Table 6** Classification Report for Gaussian for Categorical Naïve Bayes Classifier with Equal Width Discretization*

	Precision	Recall	F1-score	Support
0	1.00	1.00	1.00	11977
1	1.00	1.00	1.00	1587
Accuracy			1.00	13564
Macro avg	1.00	1.00	1.00	13564
Weighted avg	1.00	1.00	1.00	13564

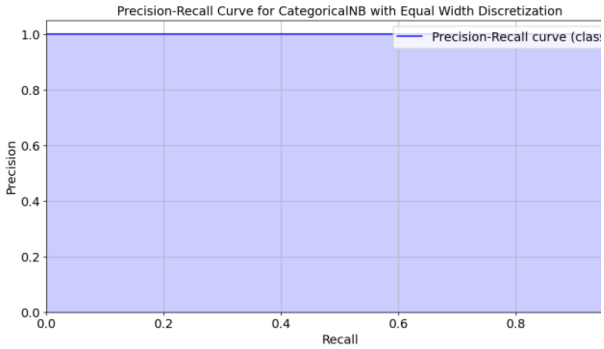

Figure 19 ***Precision-Recall Curve for Categorical Naïve Bayes Classifier with Equal Width Discretization***

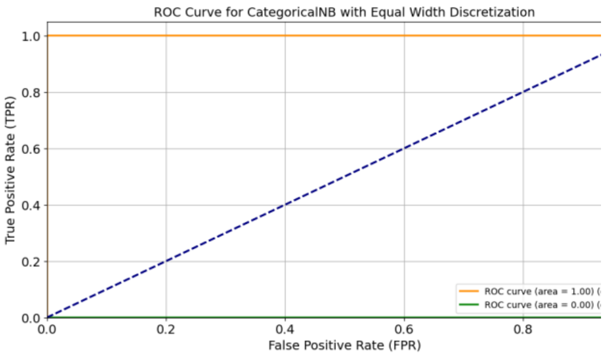

Figure 20 ***ROC Curve for Categorical Naïve Bayes Classifier with Equal Width Discretization***

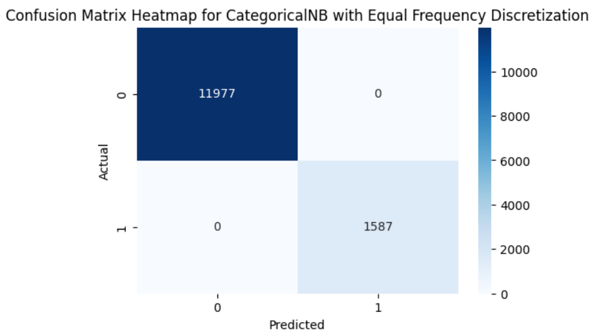

Figure 21 ***Confusion Matrix for Categorical Naïve Bayes Classifier with Equal Frequency Discretization***

Table 7 ***Classification Report for Categorical Naïve Bayes Classifier with Equal Frequency Discretization***

	Precision	Recall	F1-score	Support
0	1.00	1.00	1.00	11977
1	1.00	1.00	1.00	1587
Accuracy			1.00	13564
Macro avg	1.00	1.00	1.00	13564
Weighted avg	1.00	1.00	1.00	13564

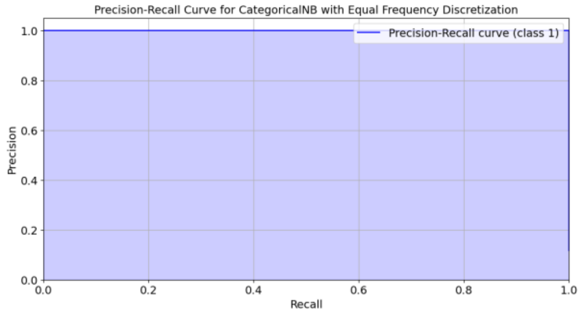

Figure 22 ***Precision-Recall Curve for Categorical Naïve Bayes Classifier with Equal Frequency Discretization***

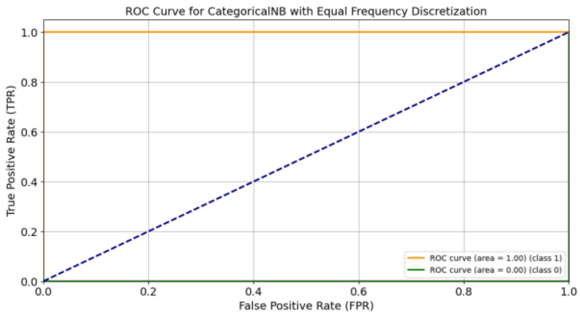

Figure 23 ***ROC Curve for Categorical Naïve Bayes Classifier with Equal Frequency Discretization***

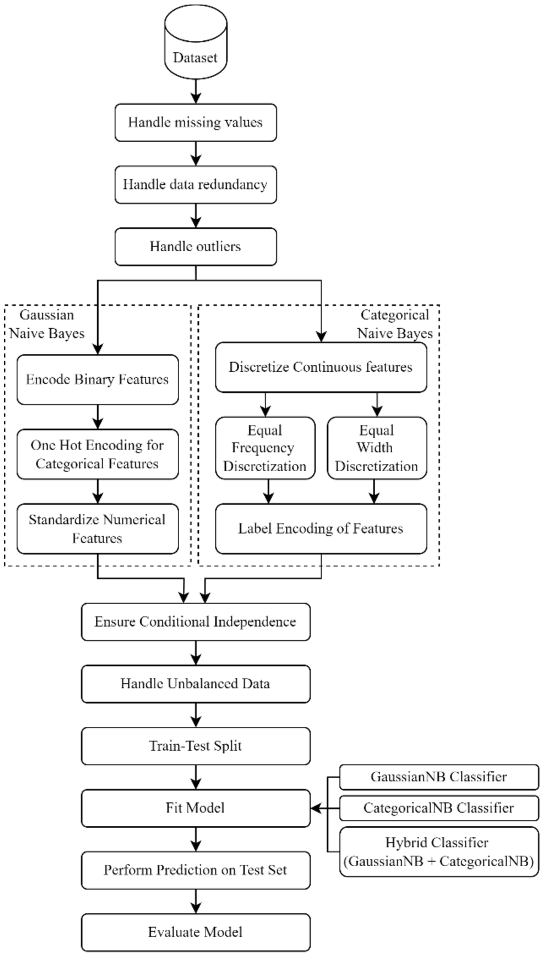

Figure 24 ***System Block Diagram***

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BIOGRAPHY

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